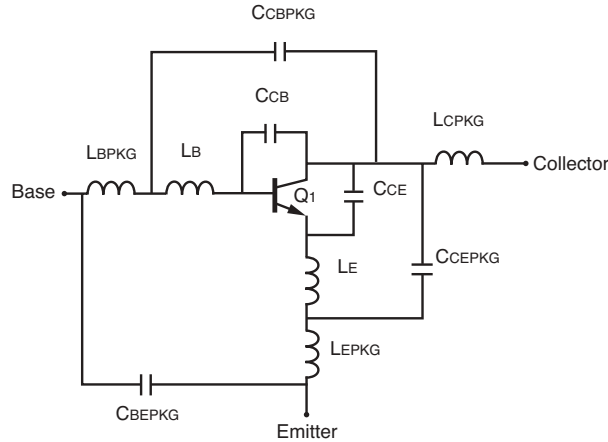


SCHEMATIC



BJT NONLINEAR MODEL PARAMETERS⁽¹⁾

Parameters	Q1	Parameters	Q1
IS	3.64e-15	MJC	0.149
BF	309.7	XCJC	1
NF	1.079	CJS	0
VA	56	VJS	0.75
IKF	233.9	MJS	0
ISE	11.67e-15	FC	0.8
NE	1.648	TF	4e-12
BR	20.01	XTF	10
NR	1.080	VTF	5
VAR	2.782	ITF	0.5
IKR	54.57e-3	PTF	20
ISC	1.024e-18	TR	0
NC	1.35	EG	1.11
RE	1.6	XTB	1.3
RB	2.2	XTI	5.2
RBM	0.05	KF	0
IRB	1e-4	AF	1
RC	4.8		
CJE	1.461e-12		
VJE	0.798		
MJE	0.137		
CJC	489.9e-15		
VJC	0.605		

(1) Gummel-Poon Model

Life Support Applications

These NEC products are not intended for use in life support devices, appliances, or systems where the malfunction of these products can reasonably be expected to result in personal injury. The customers of CEL using or selling these products for use in such applications do so at their own risk and agree to fully indemnify CEL for all damages resulting from such improper use or sale.

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DATA SUBJECT TO CHANGE WITHOUT NOTICE

ADDITIONAL PARAMETERS

Parameters	NESG2101M05
CCB	0.01 pF
CCE	0.2 pF
LB	0.16 nH
LE	0.17 nH
CCBPKG	0.45 pF
CCEPKG	0.02 pF
CBEPKG	0.05 pF
LBPKG	0.8 nH
LCPKG	1.2 nH
LEPKG	0.15 nH

MODEL TEST CONDITIONS

Frequency: 0.1 to 6 GHz

Bias: $V_{CE} = 2$ V, $I_C = 5$ mA to 40 mA

Date: 09/2003

09/02/2003

NEC

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